UKUDLA Data Science Certificate

Explanatory Analysis

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UKUDLA

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for Sustainable and Resilient Food Systems and Applied Agricultural and Food Data Science

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DAAD Deutscher Akademischer Austauschdienst German Academic Exchange Service

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Association is not yet a causal explanation

Descriptive question

- Do an exposure and outcome vary together?
- How does the association differ by group, site or period?
- Which observations support the pattern?

Causal question

- How would the outcome change under another intervention?
- Compared with which treatment or exposure condition?
- For which target units, settings and period?

An observed correlation compares different observations. A causal effect compares potential outcomes for the same target units under alternative exposures.

A model can quantify an association; assumptions give it causal meaning.

Begin causal analysis before fitting a model

Causal question and target population → Exposure contrast and outcome → Causal structure and assumptions → Identification, estimation and checks

- Define what intervention or exposure difference the effect represents.
- State which variables occur before, during and after the exposure.
- Use domain knowledge to identify common causes and selection processes.
- Decide whether the available data can identify the intended effect.

Causal analysis is a research design supported by models—not a regression command.

Define the estimand before the estimator

Target quantity

- target population;
- exposure and contrast;
- outcome and time horizon; and
- aggregation level.

Potential outcomes

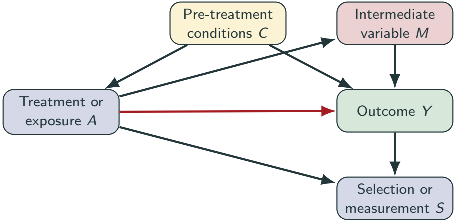
- $Y_i(a)$: outcome under condition a;
- individual effect: $Y_i(a_1) - Y_i(a_0)$; and
- average effect over target units.

Fundamental problem

- one unit experiences one realized exposure;
- the alternative outcome is unobserved; and
- identification requires valid comparisons.

"Apply treatment A" must specify its dose, timing, delivery, comparator and target unit. Otherwise the causal contrast is ambiguous.

A causal diagram makes assumptions inspectable



Use the diagram to ask

- Which variables cause both exposure and outcome?
- Which paths should adjustment block?
- Is a variable a mediator or collider?
- Which causes are measured adequately?

Arrows encode assumptions, not correlations discovered by software.

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Regression estimates a conditional association

$$Y_i = \beta_0 + \beta_1 A_i + \boldsymbol{\gamma}^{\top}\mathbf{C}_i + \varepsilon_i$$

Model components

- Y_i: outcome for target unit i;
- A_i: treatment or exposure condition;
- $\mathbf{C}_i$: selected pre-exposure covariates;
- γ: their conditional coefficients; and
- ε_i: unexplained model deviation.

Coefficient interpretation

- β_1: modeled outcome contrast per exposure contrast;
- conditional on included pre-exposure covariates;
- linear and constant unless the model says otherwise;
- causal only when identification assumptions hold.

Adding controls changes the comparison; it does not automatically remove bias.

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Identification requires assumptions beyond regression

Assumption	**Meaning**	**Questions across study designs**
Consistency	Observed outcome matches the specified exposure condition	Are protocol, dose and timing well defined?
Exchangeability	Treatment groups are comparable conditional on design or covariates	Was treatment randomized; were common causes measured?
Positivity	Relevant contrasts occur for all adjustment profiles	Does every block, site or subgroup support the contrast?
No interference	One unit's treatment does not change another unit's outcome	Can plots, samples, farms or regions affect each other?
Measurement	Exposure, outcome and confounders represent intended concepts	Are instruments, assays and reported variables valid?

Identification connects the causal estimand to the observed-data distribution. If essential assumptions are implausible, a precise estimate can remain causally uninterpretable.

Diagnostics assess the model, not the causal design

Model checks can reveal

- nonlinear exposure-outcome patterns;
- changing residual variance;
- influential observations;
- batch, block, site, temporal or grouped residual structure; and
- sensitivity to transformations and specification.

They cannot establish

- absence of unmeasured confounding;
- a well-defined intervention;
- correct causal direction;
- adequate measurement; or
- transferability beyond the target data.

Confidence intervals quantify sampling uncertainty under the model. They do not include unknown confounding, measurement error or misspecified causal structure.

Specify a causal analysis before estimation

Provisional target

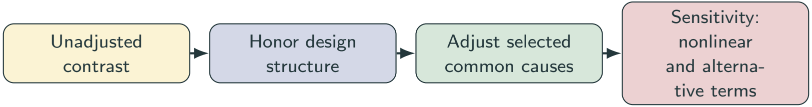
- units and target population;
- a well-defined exposure or intervention;
- outcome and timing;
- a meaningful exposure contrast; and
- target period and setting.

Required design work

- draw and justify a causal diagram;
- distinguish pre-exposure confounders from mediators;
- evaluate assignment or overlap within relevant groups;
- document unmeasured causes and exposure ambiguity; and
- decide whether the effect is identifiable.

Observed exposure variation is not automatically equivalent to a controlled intervention.

Compare estimates—then challenge their interpretation



- Report the target coefficient or contrast, uncertainty and effective n.
- Explain how the comparison changes across specifications.
- Inspect exposure overlap, residual plots, influence and temporal structure.
- Separate statistical precision from identification credibility.
- Label estimates as adjusted associations when causal assumptions remain unsupported.

Specification sensitivity is evidence to interpret—not a contest to find significance.

Key message: causal meaning comes from design

Reproducible artifacts

- causal question and estimand;
- causal diagram with justification;
- model specification table;
- coefficient and uncertainty table;
- diagnostics and sensitivity results; and
- code-linked explanatory report.

Conclusion must distinguish

- what the data describe;
- what the models estimate;
- which assumptions support identification;
- which assumptions remain doubtful; and
- which additional data or design are needed.

A credible causal analysis may conclude that the available data identify only adjusted associations. Making that boundary explicit is a scientific result, not a failure.

Estimate transparently—and earn the causal interpretation through design and assumptions.

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